

Malaria

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Malaria

Prof Onyire – Paediatrics, topic 32 · Malaria in Nigeria, 500-Level

Outline

Epidemiology & local context · Life cycle of the parasite · Pathogenesis of severe malaria · Clinical features (uncomplicated vs severe) · Diagnosis · Management of uncomplicated malaria · Management of severe malaria & complications · Malaria in pregnancy · Prevention & challenges · Mortality & take-home points · Clinical cases

Opening principle: the history is targeted at the patient in front of you – ask what matters for *this* child.

Epidemiology & local context

Nigeria carries an enormous share of the global burden: **~27% of global cases and ~31% of global deaths. Plasmodium falciparum** causes **95%** of cases and is responsible for nearly all severe disease. The vector is the **female Anopheles mosquito**. Transmission is **perennial in the South and seasonal in the North**. The susceptible (at-risk) groups are **children under 5 years, pregnant women, and non-immune adults**.

Life cycle of the parasite

Three linked stages carry the parasite from mosquito to human and back:

1. **Transmission to human** – the mosquito bites and injects **sporozoites**, which travel to the liver.
2. **Liver (pre-erythrocytic) stage** – the sporozoite infects a **hepatocyte** and multiplies.
3. **Blood (erythrocytic) stage** – the liver cell ruptures, releasing **merozoites** that infect red blood cells; inside the RBC the parasite undergoes the sexual cycle, producing forms that are taken up when another mosquito bites – completing transmission back to the mosquito.

Pre-erythrocytic stage: lasts **6-14 days** for *P. falciparum*, and there are **no symptoms** at this stage. **Erythrocytic stage:** this is where **fever and haemolysis** appear – the clinical signs.

Cyto-adherence is a special quality of *P. falciparum*: infected red cells **stick to blood-vessel walls (sequestration)**. This is one of the key causes of complications.

Pathogenesis of severe malaria

Severe disease follows a chain of events: **microvascular sequestration -> hypoxia -> lactic acidosis -> cytokine storm -> inflammatory response syndrome**. In parallel, **haemolysis** causes **anaemia and splenic dysfunction**, with **haemoglobinuria** and **thrombocytopenia**. Sequestration in the brain produces **cerebral malaria**.

Organ involvement: renal/hepatic – mild elevation of liver enzymes, jaundice, hepatic dysfunction. Respiratory – may be normal, or progress to **ARDS**. CNS/pulmonary – **pulmonary oedema** and **severe anaemia**.

Clinical features – uncomplicated vs severe

System	Uncomplicated	Severe
Fever	Intermittent; rigor; chills	High grade; continuous
CNS	Headache; malaise	Impaired consciousness / unconsciousness; seizure
Haematological	Mild anaemia; thrombocytopenia	Severe anaemia (Hb < 5); DIC
Metabolic	Mild hypoglycaemia	Hypoglycaemia; lactic acidosis

Diagnosis

1. **Clinical suspicion** – based on presentation.
2. **Rapid diagnostic test (RDT)** – detects the **HRP antigen**. Fast and good for *P. falciparum*. **Caveat: the antigen stays positive for 2-4 weeks after recovery**, so a positive RDT does not always mean current active disease.
3. **Microscopy – the gold standard**. A **thick film** gives parasite density; a **thin film** identifies the species.
4. **Parasite density** – reported per μL (e.g. $100/\mu\text{L}$ up to $100,000/\mu\text{L}$).
5. **Other investigations** – FBC, renal function, blood glucose, LFTs; **arterial blood gas** in severe disease.
6. **Level of care** – RDT at the primary health-care level; microscopy at secondary/tertiary level.

Management – uncomplicated malaria

- **ACT (artemisinin-based combination therapy)** is the treatment.
- **Antipyresis:** diclofenac or paracetamol suppository.
- **Tepid sponging** with luke-warm water to relieve fever.
- **Supportive treatment:** hydrate/infusion; blood transfusion if needed. (Artesunate should be available if the child deteriorates.)
- **Fever clearance** is expected over about **3 days**. If fever has not cleared, **look for another cause** (e.g. tonsillitis).

Reference: normal temperature of children is **37.2-37.4°C**.

Management – severe malaria & complications

Treatment:

- **IV artesunate is first-line** – continue until the child regains consciousness.
- **Second line: quinine.**
- **Calculate the plasma parasite density.**

Management of complications:

- **Airway** – keep it patent.
- **Convulsions** – give an anticonvulsant.
- **Avoid steroids.**
- **Severe anaemia** – transfuse blood, especially if the child is in distress; **transfuse if Hb < 5 g/dl**.
- **AKI** – watch fluid balance.
- **Hypoglycaemia** – check blood glucose **4-hourly** and correct with **10% dextrose**.

Rule of thumb: Hb x PCV. Sickle-cell patients need a stable Hb kept where they are comfortable.

Malaria in pregnancy

- **1st trimester: quinine + clindamycin.**
- **Other trimesters: ACT.**

Prevention & challenges

Prevention strategies: vector control (insecticide; avoid bushy environments) and **chemoprophylaxis** (Fansidar in pregnancy; mefloquine, doxycycline).

Challenges: drug resistance in *P. falciparum*, including **partial resistance to ACT**; **RDT false negatives**; **access to care**; and poor **antimicrobial stewardship** (abuse of antimicrobial drugs).

Mortality & take-home points

Even when treated, mortality in severe malaria can be high – **fast treatment is essential**.

Take-home: severe malaria with **fever + unconsciousness** -> **think cerebral malaria**.

Clinical cases

Case 1 (severe malaria). A 14-year-old male with 4-day fever, headache and rigor; Hb 6.2 g; RDT positive; parasite density > 100,000/μL. *Why is IV artesunate better than quinine?* – Quinine risks arrhythmias; artesunate gives **faster parasite clearance, lower mortality**, and a **better safety profile** (quinine can cause hypotension and hypoglycaemia).

Case 2 (emergency). A 4-year-old with fever and seizure; RDT positive but blood film negative. This is an **emergency – severe malaria until proven otherwise**. The RDT shows they have been infected; check for another cause (FBC, LFT, etc.) and **repeat microscopy in 12-24 hours**.

Assignment: why does severe malaria cause hypoglycaemia, and why is it dangerous in a child?

References

Lecture material of Prof Onyire (500-level Paediatrics, Abia State University); WHO malaria treatment guidelines; Nelson Textbook of Paediatrics.